

LAKSHMI OMKARESWAR THUMMAGUNTA

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OBJECTIVE

ECE undergraduate with Honours in VLSI Design, skilled in RTL design, FPGA development, and embedded systems, with growing expertise in cyber security. Driven to engineer efficient, secure hardware solutions and contribute meaningfully to cutting-edge ASIC or embedded systems teams.

EDUCATION

B.E in Electronics and Communication Engineering, SCSVMV (Deemed to be University)
Honours in VLSI Design
CGPA – 8.74/10.00

2024 – 2027
Kanchipuram, TN

Diploma in Electronics and Communication Engineering,
Audisankara College of Engineering & Technology

2021 – 2024
Nellore, AP

TECHNICAL SKILLS

- Programming Languages:** C, C++, Python, Embedded C, VHDL, Verilog HDL, System Verilog, MATLAB
- Design:** FSM Design, RTL Design, Digital Electronics, FPGA Design, ASIC Design, UART/SPI/I2C, GPIO
- Embedded Systems:** ESP32, 8051 Microcontroller, Hardware Interfacing, Sensor Integration
- Tools:** Xilinx Vivado, LT-Spice, P-Spice, ModelSim, Multisim, GTKWave, MATLAB, CC Studio, VS Code, Arduino IDE, IAR Embedded Workbench, CISCO Packet, Wireshark
- Domains:** VLSI Design, Embedded Systems, IoT Development, Wireless Communication

PROFESSIONAL EXPERIENCE

Cyber Security Intern, Soprol7 Analytics Private Limited

- Monitoring network traffic and security events using SIEM tools to detect anomalies, suspicious activity, and potential threats in real time.
- Conducting security audits and assessments to identify vulnerabilities across systems and applications, and documenting findings with actionable remediation recommendations.
- Assisting in the preparation of security audit reports and compliance documentation to support organizational risk management processes.

06/2026 – 07/2026
Remote, India

Internship Trainee, Andhra Pradesh Power Generation Corporation Limited

- Studied real-time operations of a 800 MW power generation unit, including turbine and electrical distribution systems, supported efficiency analysis and daily performance data logging, contributing to maintenance decisions
- Collaborated with industry professionals to strengthen problem-solving abilities and gain practical exposure to power systems, energy management, and technical analysis.

11/2023 – 05/2024
Nellore, Andhra Pradesh, India

PROJECTS

AI-Based Automatic Bike Cleaning System

- Designed FSM-based automated control system using sensor inputs and microcontroller.
- Implemented real-time hardware interfacing (motors, pumps, sensors) via GPIO.
- Finalist – InnoWAH (PALS).

11/2025 – 03/2026

Mini RISC Processor | Verilog HDL

- Designed and simulated an 8-bit single-cycle RISC processor using Verilog HDL with ALU, register file, control unit, and memory interface in Vivado.

09/2025 – 10/2025

AI-Powered OLED Display Interface ESP32

- Created a cutting-edge embedded system that combines OpenAI API with a 128x64 OLED display, employing efficient I2C communication along with optimized memory and power management for enhanced edge-AI applications.

04/2025 – 04/2025

CERTIFICATIONS & RECOGNITIONS

VLSI Design – Internshala

- Completed an 8-week VLSI Design training covering HDL fundamentals, combinational & sequential logic, FSMs, and FPGA based system design

Computer Networks and Internet Protocol – NPTEL

- Covered core networking concepts including the OSI/TCP-IP models, routing protocols, and internet communication standards.

Finalist, InnoWAH Challenge (PALS) – Recognized for designing a smart, sensor-driven automated system with real-world impact.

Special Prize for Innovative Engineering Solution, Think2Impact (Hackathon) – Awarded for presenting and validating the project in a competitive, industry-oriented evaluation round.